

AAHIL RUPSI

Software Engineer | Atlanta, GA

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EDUCATION

University of California, San Diego (UC San Diego)

June 2025

B.S. in Mathematics—Computer Science, Minor in Data Science

La Jolla, CA

Relevant Coursework: Algorithm Design & Analysis, Artificial Intelligence, Machine Learning, Operating Systems, Database Systems, Linear Algebra, Probability & Statistics, Discrete Mathematics

EXPERIENCE

Software Engineer

October 2025 – August 2026

Mighty Investment Group (Parent fund; supporting 2 portfolio companies concurrently)

Tucker, GA

- Built a secure investor portal (**Svelte**, JWT auth) giving **50+** investors self-service access to portfolio performance and fund documents across **\$200M+** AUM, replacing manual, email-based reporting
- Built an in-house AI retail void/gap-analysis engine, auto-scoring site-selection opportunities across **14** retail categories, encoding the firm's proprietary underwriting criteria unavailable in commodity tools like SiteSeer

McKinney & Co. | Commercial Insurance Vertical (\$100M book of business)

- Led a **4-person** offshore engineering team as sole US liaison to shareholders, delivering a unified **Next.js** AI portal that consolidated **6+** internal tools into one workflow app used daily by **30+** staff from lead intake to bind
- Built an AI quality-control pipeline that scrapes all transaction documents throughout the quoting process and cross-references them via LLM extraction to ensure zero discrepancies, catching errors across **150+** policies/month and reducing E&O exposure by **~30%**
- Developed an automated email-processing pipeline that parses incoming messages, files them under the corresponding business directory, and indexes the content to allow an AI agent to query the records during quality-control audits
- Built **Python/Playwright** RPA automations propagating policy data across **5+** downstream systems, cutting duplicate manual entry by **~15 hrs/week** and errors by **~90%**
- Built a **Python** voice AI receptionist using the **Gemini Multimodal Live API** over WebSockets and integrated with **OpenPhone** IVR to triage **~500** inbound calls/month, cutting front-desk call load by **~40%**
- Launched **InsureCStore**, a **React** SPA delivering the first-of-its-kind instant online quoting experience for commercial insurance (a capability no other carrier or agency offers), generating **50+** leads/month

Cinco Auto Insurance | Personal Auto Vertical (250K+ monthly visitors)

- Single-handedly migrated cincoautoinsurance.com from a legacy PHP stack (**7,823** files, no CI/CD) to **AstroJS**, **Cloudflare Workers**, and **D1**, consolidating to **~200** files while regenerating **5,700+** bilingual pages
- Built a custom headless CMS (**AstroJS SSR + D1**) letting non-technical staff publish pages and blog content directly, with automatic edge-cache invalidation, removing developers from the publishing loop
- Re-architected the site around edge-first rendering (**Cloudflare Workers**, **D1**, **R2**, automatic image transformation) with automated CI/CD (**GitHub Actions**), sustaining **hundreds of thousands** of monthly visitors at high Lighthouse scores
- Set up scheduled **GitHub Actions** runners to pipe the company's Agency Management System (AMS) records into **Google BigQuery**, enabling self-serve analytics beyond AMS's built-in reporting

IT Service Desk Technician

September 2024 – September 2025

UC San Diego, Information Technology Services (ITS)

La Jolla, CA

- Resolved Tier 1 IT tickets (account recovery, VPN/Wi-Fi configuration, hardware/software troubleshooting) for **75,000+** students, faculty, and staff via UCSD's **ServiceNow** platform
- Built a RAG pipeline and Tier 2/3 escalation routing system (**~85%** accuracy) for a Chrome extension assistant, embedding **10,000+** historical tickets (**OpenAI**, **FAISS**) behind a Flask/AWS Lambda API to surface resolutions in real time, cutting resolution times by **~25%**

PROJECTS

MNIST Digit Recognizer | *Rust*, *WebAssembly* | [GitHub](#)

May 2026 – August 2026

- Implemented a neural network ($784 \rightarrow 128 \text{ ReLU} \rightarrow 10 \text{ softmax}$) from scratch in Rust, hand-writing forward/backward propagation and mini-batch SGD, reaching **~98%** test accuracy on all 60,000 MNIST images
- Compiled the model to **WebAssembly** via wasm-pack into an **850KB** binary, running the identical Rust forward pass client-side in under **100 μ s** per prediction with zero server calls
- Built a canvas drawing UI deployed at mnist-wasm.aahilrupsi.com, letting users sketch digits and see live predictions with per-class confidence